# Nosocomial Infections

## A History of Hospital-Acquired Infections

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**KEYWORDS**

• Health care acquired • Nosocomial • Pneumonia • Urinary tract • Bloodstream
• Surgical site

**KEY POINTS**

- Sixth leading cause of death in the United States.
- 99,000 annual deaths.
- Contributes to multidrug-resistant organisms.
- Most infections are preventable.

## INTRODUCTION

In the United States, health-care acquired infections (HAIs) or nosocomial infections (NIs) are the sixth leading cause of death, surpassing the combined deaths from human immunodeficiency virus/AIDS, cancer, and traffic accidents.[1,2] Since Hippocrates's time, physicians have been aware HAIs harm patients.[1] HAIs refer to infections that arise in any inpatient or outpatient setting and appear 48 hours after hospitalization, or within 30 days after receiving health care, or up to 90 days after undergoing certain surgical procedures.[1,3,4] The highest at-risk population are patients who are immunocompromised, such as transplant patients, chemotherapy patients, and neonates, and critically ill admitted to burn units or intensive care units (ICUs).[5] In 2002, there were 1.7 million NIs, approximately 99,000 hospitalized patients who died of NIs, and 1 in 31 hospitalized patients per day developed an NI.[2,6,7] The direct economic burden of HAIs in the United States is estimated to be between $28 and $34 billion, of which $25 to $32 billion could be prevented with effective infection-control programs.[8]

To combat and reduce HAIs, the Centers for Disease Control and Prevention (CDC) National Healthcare Safety Network (NHSN) and the Centers for Medicare and

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Medicaid Services implemented programs to monitor and reduce the rates of NIs.[9] The NHSN is now the largest tracking system of HAIs in the United States and includes 25,000 medical facilities spanning across 50 states and US territories.[10,11] This is an improvement compared with the mid-1970s, when only 62 US hospitals participated in programs to reduce HAIs.[11,12]

Catheter-associated urinary tract infections (CAUTIs), hospital-acquired pneumonias (HAPs), bloodstream infections (BSIs), and surgical site infections (SSIs) cause most HAIs.[5] Among them, CAUTIs are the most prevalent worldwide NIs, causing 40% of all US NIs or 900,000 cases annually.[5,13] However, the most deadly NIs are HAPs and BSIs, which together account for 67% of US annual deaths associated with NIs.[14] Lower tract urinary infections are often the precursor for upper tract urinary infections and BSIs. Urinary tract infections are the leading cause of secondary bacteremia and account for 17% of nosocomial bacteremias.[13,15]

A significant factor in the rising mortality of NIs is the increasing prevalence of multidrug-resistant organisms (MDROs), which can render classes of antibiotics ineffective in treating many common infectious diseases. Annually more than 2 million people acquire an antibiotic-resistant infection, and of those, 23,000 die.[12,16] These organisms can survive on inanimate surfaces for prolonged periods, contaminate hospital rooms and equipment, and are spread by the hands of health care workers.[17–20] ICU patients are disproportionately affected because of chronic comorbid diseases, more aggressive interventional treatments with indwelling devices, and frequent exposure to antibiotics. The mortality rate of ICU patients is estimated to be more than twice that of non-ICU patients.[21] Aerobic gram-negative bacteria (GNB) account for 40% of all ICU deaths and 70% of all ICU HAIs.[22] GNB have become the most drug resistant because of lack of development of new antimicrobials. The last novel class of antibiotics against GNB was released in 1996: the fluoroquinolones. Several risk factors are associated with acquiring MDROs including prior antibiotic use, septic shock, and 5 or more days of hospitalization.[23]

In this article we focus on the history, prevalence, economic costs, morbidity and mortality, along with risk factors of the most common HAIs; address the emergence of MDROs, including *Staphylococcus aureus* (SA) and *Clostridium difficile*; and discuss examples of effective programs that can reduce infection risks.

## NOSOCOMIAL URINARY TRACT INFECTIONS

Nosocomial urinary tract infections are the most common HAIs reported to the NHSN, European Center for Disease Prevention and Control, and the World Health Organization.[24] Most nosocomial urinary tract infections are caused by instrumentation of the urinary tract with either urethral or suprapubic catheters.[13,24] CAUTIs account for 75% of HAIs.[13] Between 15% and 25% of hospitalized patients have indwelling urinary catheters inserted during their hospitalization[13] and many are often inappropriately placed.[25] The highest risk factor for developing CAUTIs is prolonged catheter use.[13,26] Catheter colonization increases the risk of bacteriuria by 3% to 10% per day.[13] After 30 days, 100% of catheters become colonized with a biofilm, where organisms exist in a protected state from antimicrobials and host defenses. The only effective eradication is removal of the catheter.[24,27]

Diagnosis of CAUTIs requires the presence of an indwelling or suprapubic catheter plus infectious signs or symptoms and a positive urine culture. Among patients who develop bacteriuria only 10% to 25% are symptomatic.[24,28] Asymptomatic bacteriuria in catheterized patients, even in the presence of malodorous, cloudy urine and bacteria growth, do not have CAUTI and do not require antimicrobial therapy.[26]

Over the past few years, NHSN reported declining use of urinary catheters, which has reduced CAUTI rate. Prevention strategies included decreasing unnecessary catheter placement, limiting the duration of catheterization, securing closed drainage systems, and encouraging aseptic technique with only trained personnel performing catheter insertion. Figs. 1 and 2 demonstrate improvement over the past several years in CAUTI rates.[9]

## HEALTH CARE–ASSOCIATED PNEUMONIAS

Health care–associated pneumonias (HCAPs) are the second most common HAI; however, it is the most deadly.[22,23,29] Every 5 to 10 hospitalized patients out of 1000 develop HCAP. Patients with the highest risks of acquiring HCAPs include infants; young children; adults older than 65 years of age; and patients who are ventilated, immunosuppressed, have decreased sensorium, recent thoracoabdominal surgery, and cardiopulmonary disease.[29] HCAPs, which cause 86% of NIs, are divided into HAPs, which occur after 48 hours of admission, and ventilator-acquired pneumonias.[14] The latter causes 86% of all HCAPs.[5] Between 9% and 27% of mechanically assisted patients are at risk for ventilator-acquired pneumonia within 48 hours after endotracheal intubation.[1,5] HCAP account for 22% of all ICU NIs and are directly associated with prolonged hospitalizations, rising costs, and increased mortality and morbidity.[1,14,23,30] MDROs frequently cause ventilator-acquired pneumonia, which is also a major factor in morbidity and mortality.[23] There are several methods to prevent HCAP, including pneumococcal vaccination, use of incentive spirometry, and chest physiotherapy.[29]

The diagnosis for HCAPs is challenging in patients with comorbid conditions and requires documentation of multiple nonspecific clinical factors including fever, cough, productive sputum, and radiologic evidence of an infiltrate.[23] HCAP pathogens are often the intrinsic colonizing oral flora of patients. The natural human oral flora is

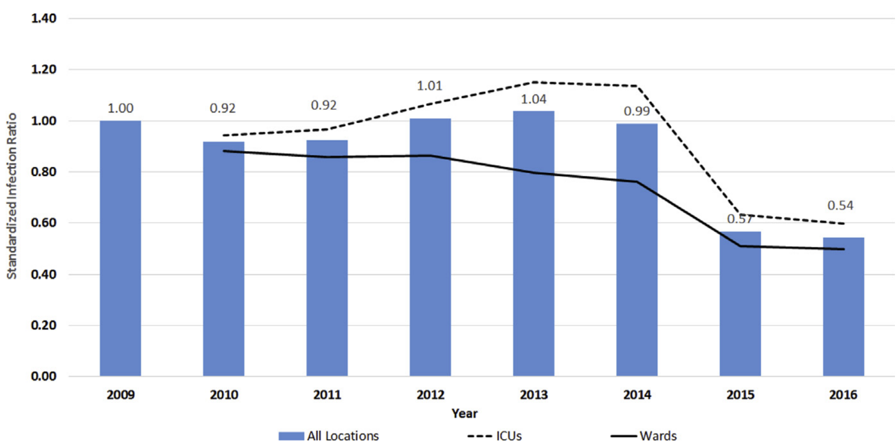


**Fig. 1.** Changes over time in CAUTI standardized infection ratios in US hospitals using 2009 baseline, NHSN 2009 to 2016. SIR, standardized infection ratios. (*Data from*: Centers for Disease Control and Prevention. Data Summary of HAIs in the US: Assessing Progress 2006-2016. https://www.cdc.gov/hai/data/archive/data-summary-assessing-progress.html?CDC_AA_refVal=https%3A%2F%2Fwww.cdc.gov%2Fhai%2Fsurveillance%2Fdata-reports%2Fdata-summary-assessing-progress.html. Updated December 5, 2017. Accessed September 18, 2019.)

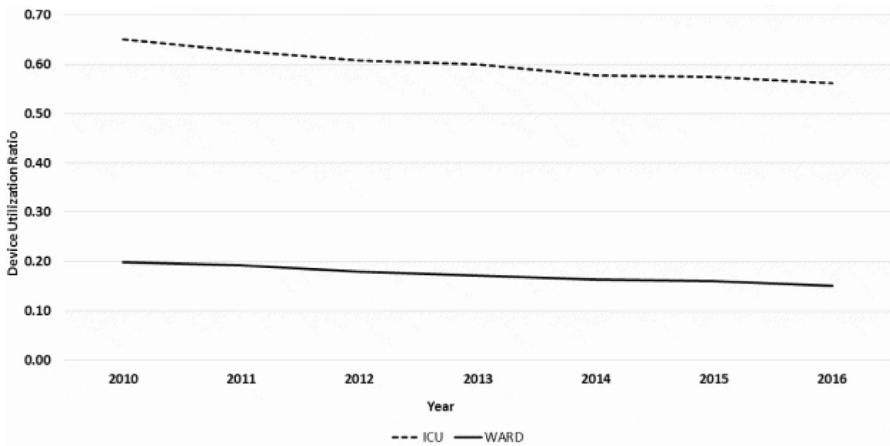


**Fig. 2.** Changes over time catheter use ratio in US hospitals, 2010 to 2016. (*Data from*: Centers for Disease Control and Prevention. Data Summary of HAIs in the US: Assessing Progress 2006-2016. https://www.cdc.gov/hai/data/archive/data-summary-assessing-progress.html?CDC_AA_refVal=https%3A%2F%2Fwww.cdc.gov%2Fhai%2Fsurveillance%2Fdata-reports%2Fdata-summary-assessing-progress.html. Updated December 5, 2017. Accessed September 18, 2019.)

composed of a healthy mixed population of gram-positive, gram-negative, and anaerobic bacteria, which changes because of several factors including age, chronic illness, orthodontic procedures, antibiotic exposure, and hospitalization.[23,31] Aerobic GNB account for greater than 73% of HCAP cases and frequently respiratory samples are polymicrobial.[23,29]

Interventions minimizing HCAPs include preventing aspiration, sterilization of respiratory devices, aggressive hand hygiene, and reducing GNB colonization through oral care.[19] In 2017, NHSN showed a 5% statistically significant decrease in ventilator-associated events, which was an improvement from 2015 to 2016 data that showed a 2% decrease.[9]

## NOSOCOMIAL BLOODSTREAM INFECTIONS

The incidence of nosocomial BSIs between 1995 and 2002 was 60 cases per 10,000 hospital admissions. Intravascular devices, of which 72% were central venous catheters, were the most common predisposing factor.[32] NHSN reported between 2011 and 2014 there were 85,994 central line–associated bloodstream infection (CLABSIs) with a mortality rate of 10% to 20%.[33] In 2000, the economic cost was $40,000 per bacteremic survivor.[34]

The potential infectious breaches on a central venous catheter are demonstrated in **Fig. 3**.[33,35] Risk factors for CLABSIs include: site of insertion, multiple lumen catheters, total parenteral nutrition, low nurse-to-patient ratio, secondary infections, intravenous catheterization for longer than 3 days, inexperienced personnel, and use of stopcocks.[33] Any organism penetrating a central venous catheter can cause CLABSIs. The most common organisms are *Staphylococcus* species, Enterococci, *Candida* species, and GNB infections.[36] Drug resistance, including fluoroquinolone resistance and extended-spectrum β-lactamase-producing organisms is becoming more prevalent. Carbapenem-resistant Enterobacteriaceae pathogens comprised 7.1% of

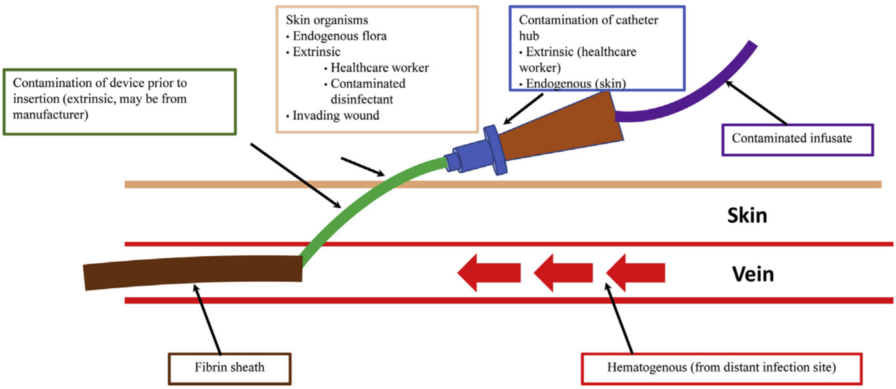


**Fig. 3.** Sources of infection from central venous catheter. (*Data from*: Agency for Heatlhcare Research and Quality. Appendix 3. Guidelines to Prevent Central Line-Associated Blood Stream Infections. U.S. Department of Health and Human Services. https://www.ahrq.gov/hai/clabsi-tools/appendix-3.html#s14. Updated March 2018. Accessed September 18, 2019; and Safdar N, Maki DG. The pathogenesis of catheter-related bloodstream infection with noncuffed short-term central venous catheters. *Intensive Care Med.* 2004;30(1):62-67.)

CLABSIs reported in 2014.[36] Measures to prevent CLABSIs include: stringent infection control practices, use of chlorhexidine as skin preparation before line insertion, removal of unnecessary catheters, and avoidance of femoral vein catheterization.[33,37]

Between 2008 and 2016, there was a 50% national reduction in CLABSIs as a result of collaborative efforts between national and state agencies.[9] **Fig. 4** illustrates CLABSIs standardized infection ratios calculated using 2015 and historical baseline model, and demonstrates the changes in CLABSI in US hospitals between 2006 and 2016.[9]

With data now helping guide best practices for catheter insertion and care, rates of CLABSIs reported by NHSN continue to decline. Between 2016 and 2017 there was a 9% statistically significant decrease in CLABSIs.[9]

## NOSOCOMIAL SURGICAL SITE INFECTIONS

Nosocomial SSIs (NSSIs) are the most frequent unplanned cause for postoperative readmission[38] and the most common NI among surgical patients.[39] NSSIs are also the most costly HAI type.[40] NSSIs, which causes 11% of all ICU deaths,[2] are classified by the CDC to be an infection in an “incisional, organ or organ space manipulated during an operation” that occurs within 30 days of surgery or within 90 days of surgery if a prosthetic device is implanted.[41] In 2006, 80 million surgical procedures were performed in the United States of which 2% to 5% developed NSSIs.[5,40] This estimate is believed to be underreported because approximately 50% to 84% of NSSIs are evident after discharge.[2] In addition, certain techniques are used to achieve lower NSSI rates and reporting is also voluntary.[2,39] The impact of NSSIs include poorer patient outcome, increased hospital stays, and lower reimbursements to hospitals.[39] Each NSSI may cost from $1000 to $90,000 to treat. In the United States, the total annual cost of NSSIs is estimated to be $700 million.[8]

The incidence of NSSI varies from surgical procedure, surgeon, other health care providers, surgical center, and patient. Patients at highest risk for NSSIs are older in age with an emergent admission and with at least one comorbidity.[42] Approximately half of NSSIs are estimated to be preventable. Modifiable risks for SSIs include:

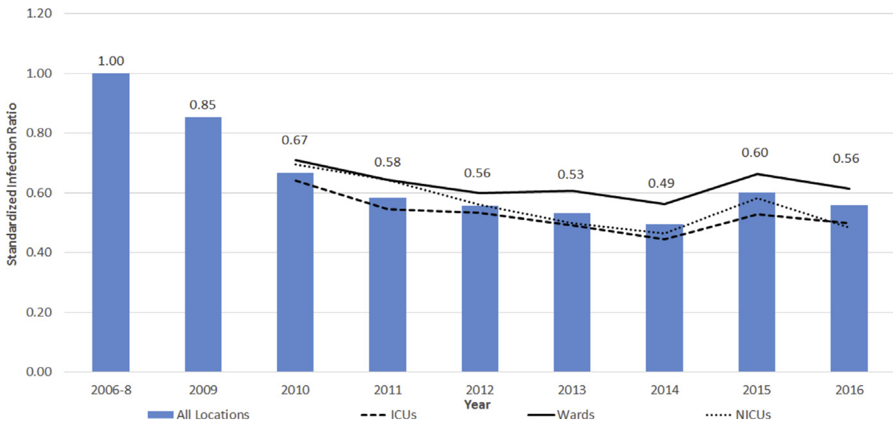


**Fig. 4.** Changes over time in CLABSI standardized infection ratios in US hospitals using 2006 to 2008 baseline, NHSN 2009 to 2016. NICU, neonatal ICU. (*Data from*: Centers for Disease Control and Prevention. Data Summary of HAIs in the US: Assessing Progress 2006-2016. https://www.cdc.gov/hai/data/archive/data-summary-assessing-progress.html?CDC_AA_refVal=https%3A%2F%2Fwww.cdc.gov%2Fhai%2Fsurveillance%2Fdata-reports%2Fdata-summary-assessing-progress.html.)

minimizing presurgical risk factors (smoking cessation, malnourished state), immunosuppressed state, appropriate surgical preparation, and optimizing conditions during surgery.[4,39] The CDC monitors several surgeries for NSSIs, including cardiovascular, colon, rectal, pelvic, hip, and knee procedures.[40]

NSSIs are commonly caused by bacteria or fungi invasion of the skin, soft tissue, or organ space via trauma or surgery.[43] For clean surgeries that do not involve the gastrointestinal, respiratory, or female genital tracts, gram-positive bacteria (GPB), such as SA, coagulase-negative staphylococcus, and streptococcal species from the endogenous or exogenous environment are the most common organisms cultured. For surgeries involving the gastrointestinal or female genital tracts, in addition to GPB, GNB coliforms and anaerobes need to be considered. To optimize bacterial growth, cultures should be obtained before initiating antibiotics to identify the causative organism and narrow antimicrobial coverage.

A breach in sterile technique from any health care provider can contribute to NSSIs. In the 1990s, propofol vials were reused and contaminated by anesthesia personal, which resulted in 155 patients infected across 20 states and of those four patients died. Isolated organisms included GPB, GNB, and fungi.[44]

NHSN reported an overall decrease in NSSIs from 2010 to 2014 as illustrated in **Fig. 5**.[9] Between 2016 and 2017 there was a 1% statistically significant improvement in 10 procedures but there was no improvement in abdominal hysterectomy or colon surgery SSIs.[9]

## OUTBREAKS

Outbreaks of HAIs can be with a singular organism or polymicrobial and etiologies have included bacteria, fungi, viruses, and prions. A variety of reservoirs including contaminated medical equipment, such as duodenoscopes, reused injectable solutions, and health care workers have been reservoirs. Various outbreaks over the last 20 years are listed in **Tables 1** and **2**.[4,45–55] Methods to prevent and control outbreaks

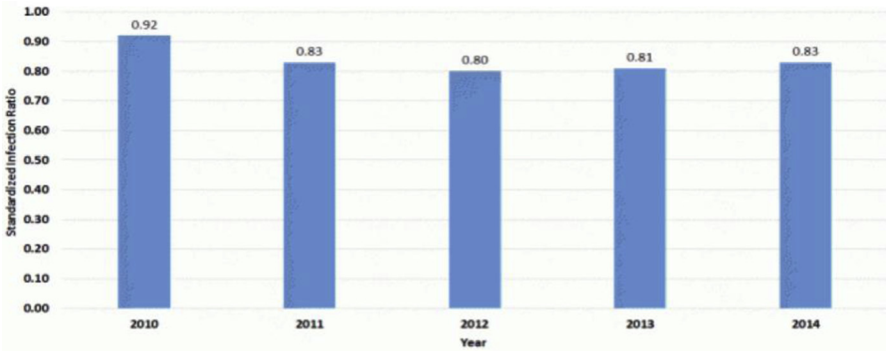


**Fig. 5.** Changes over time in SSI after any of 10 surgical care improvement project procedures in US hospitals using 2006 to 2008 baseline, NHSN 2010 to 2014. (*Data from*: Centers for Disease Control and Prevention. Data Summary of HAIs in the US: Assessing Progress 2006-2016. https://www.cdc.gov/hai/data/archive/data-summary-assessing-progress.html?CDC_AA_refVal=https%3A%2F%2Fwww.cdc.gov%2Fhai%2Fsurveillance%2Fdata-reports%2Fdata-summary-assessing-progress.html. Updated December 5, 2017. Accessed September 18, 2019.)

**Table 1**
**Summary of nosocomial outbreaks**

| Year | Organism | Location | Number of Cases | Number of Deaths | Source |
|---|---|---|---|---|---|
| 2016 | *Candida auris* | ICU in United Kingdom | 70 | 0 | Reusable axillary temperature probes[45] |
| 2016 | *Bulkholderia cepacia* | 59 long-term and rehabilitation facilities[a] | 163 | 9 | 0.9% sodium chloride IV flush by nurse assistant[46] |
| 2016 | Polymicrobial[b] | Quebec, Canada | 9 | 0 | Suboptimal cavitron ultrasonic surgical aspirator cleaning used in craniotomy/ craniectomy.[54] |
| 2014 | Carbapenemase-resistant Enterob-acteriaciae | UCLA | 9 | 2 | Duodenoscopes[47] |
| 2012–2013 | *Exserohilum rostratum* | 25 states | 753 | 64 | New England Compounding Center[48] Preservative-free MPA |

*Abbreviations:* IV, intravenous; MPA, methylprednisolone acetate; UCLA, University of California, Los Angeles.

[a] Delaware, Maryland, New Jersey, and Pennsylvania.

[b] *Opionibacterium acnes* (n = 5), *Staphylococcus aureus* (n = 2), *Streptococcus agalactiae* (n = l), *Enterococcus faecalis* (n = l), *Bacteroides fragilis* (n = l), polymicrobial (n = 2).

**Table 2**
**Summary of nosocomial outbreaks**

| Year | Organism | Location | Number of Cases | Number of Deaths | Source |
|---|---|---|---|---|---|
| 2011 | *Klebsiella pneumoniae* carbapenemases | US NIHCC-ICU[a] | 18 | 11 | Inanimate objects Ventilators Hospital equipment[49] |
| 2008 | MRSA and MSSA (FusR) | United Kingdom[b] | 19 | | MSSA (FusR) colonized health care worker with psoriasis[50] |
| 2008 | HCV | New York[c] | 657 exposed, 3 seroconverted | 0 | Hemodialysis machine[51] |
| 2006–2007 | HCV and HBV | New York colonoscopy clinic | 50,000 exposed, 6 HBV, 6 HCV, 1 HBV + HCV | 0 | Reused syringes administering propofol vials by 1 anesthesiologist[52] |
| 1998 | HCV | Florida | 3 | 0 | Reused multisaline vials[55] |
| 1998 | Pseudomonas UTI | Pediatric surgical ICU | 14 | 0 | Tap water[53] |

*Abbreviations:* HBV, hepatitis B virus; HCV, hepatitis C virus; MRSA, methicillin-resistant *Staphylococcus aureus*; MSSA, methicillin-sensitive *Staphylococcus aureus*; UTI, urinary tract infection.

[a] US National Institutes of Health Clinical Center-ICU.
[b] Nottingham University Hospital, Nottingham, UK.
[c] New York Outpatient Hemodialysis Center.

include strict handwashing, proper cleaning of equipment, isolation of patients colonized with MDROs, and antimicrobial stewardship.[56]

### *Staphylococcus aureus*

SA is a highly adaptable and virulent organism and is among the most common causes of NIs, including CLABSI, infective endocarditis, HCAP, skin and soft tissue infections, and SSIs.[57] Approximately 30% of people are asymptomatic carriers in the back of their noses.[58] Exogenous spread of SA occurs by direct contact with an infected wound or from contaminated health care workers' hands.[59]

SA was initially treated with penicillin; however, resistance occurred in hospitals shortly after its regular use.[60–62] To overcome SA's penicillin resistance, methicillin, a semisynthetic penicillin, was introduced in 1959, but methicillin-resistant SA (MRSA) emerged shortly thereafter in 1961. The first reported US outbreak of hospital-acquired MRSA infection occurred in 1963.[60,61] Over time, the rate of hospital-acquired MRSA in US hospitals increased from 4% in the 1980s to 50% by the late 1990s.[59,60] However, community-acquired MRSA infections were sporadic in nature until the 1990s, when community-acquired MRSA infections by a new strain (USA 300 clone) were reported among children,[63] team members,[64,65] correctional facilities,[66] day care centers,[67] and military personnel.[68,69] Community-acquired MRSA strains subsequently integrated in nosocomial settings.[60] By 2013, the CDC deemed MRSA to be a serious threat to the world.[16] Risk factors for MRSA infection include immunocompromised state, prosthetic or invasive devices, residing in the ICU, MRSA self-colonization, and open wounds.[70]

By 2011, invasive MRSA infections are estimated to have caused 80,000 US cases and 11,000 annual deaths.[16] Between 2005 and 2014, the overall incidence of hospital-acquired MRSA infections declined by 65%, contributing to interventions designed to decrease transmission and reduce the risk of device- and procedure-associated infections.[59,71] CDC's most recent report on HAIs demonstrates an 8% statistically significant reduction in MRSA bacteremia between 2016 and 2017.[7]

Since its introduction in 1972, vancomycin became the only reliable agent against MRSA. Vancomycin-intermediate SA (minimum inhibitory concentration of 4–8 $\mu g\ mL^{-1}$), first reported in 1997, and vancomycin-resistant *S aureus*, first described in Japan in 1996,[72] both entered the United States in 2002[73] and have become emerging threats to the world.[74] Risk factors for vancomycin-resistant *S aureus* include MRSA and vancomycin-resistant enterococci colonization, MRSA bacteremia with central line or prosthesis, chronic medical conditions (diabetes, dialysis), prolonged vancomycin use, and chronic skin ulcers.[75,76]

### *Hepatitis C Virus*

Nosocomial hepatitis C virus (HCV) infection was historically most commonly transmitted via hemodialysis (HD), blood transfusions, and needle stick injuries. However there have been reports of other mechanisms of transmission including unsafe injection practices with reused contaminated medications, inadequate cleaning of reusable contaminated equipment, and colonoscopy with biopsies.[52]

Since the advent of blood donation screening for HCV in 1990, the risk of HCV in blood transfusion has been nearly eliminated. Before 1990, 10% of transfusion recipients acquired HCV.[77] Currently, HD still poses the highest risk, especially in-hospital HD. Periodic screening is recommended for HD patients because in some parts of the United States, almost 10% of HD patients become seropositive. There are several factors that increase a patient's risk during HD, including the prevalence of HCV patients in the dialysis units and length of time on HD.

### *Clostridium difficile* Infection

The bacteria *C difficile* was first isolated from the stool of a healthy infant in 1935; however, pseudomembranous colitis was first described in 1893. In 1978, following the introduction of broad-spectrum antibiotics in the 1960s and 1970s, *C difficile* was determined to be the organism responsible for most cases of antibiotic-associated diarrhea and pseudomembranous colitis.[78]

A new *C difficile* strain highly resistant to clindamycin, which was discovered between 1989 and 1992, was identified to have caused *C difficile* infectious (CDIs) outbreaks in four unrelated US hospitals.[79] By the early 2000s several US and Canadian institutions noted an increasing incidence and severity of CDIs among patients exposed to fluoroquinolones and cephalosporins.[80–82] Between 2000 and 2003 *C difficile* isolates collected from eight health care facilities in six states discovered a highly transmissible, more resistant and virulent strain termed BI/NAP1/027,[83] which produced about 10-fold more toxin A and 23-fold more toxin B than previous strains.[84,85] Higher relapse rates also occurred.[86]

Currently, CDIs cause HAIs worldwide and more than a million cases per year in the United States.[87] Between 10% and 30% of patients develop at least one recurrent CDI after the first diagnosis. The risk of recurrence increases with each successive episode.[88,89] Recurrent CDI is associated with a 33% increased risk of mortality at 180 days relative to patients who do not develop a recurrence.[89] In 2013, the CDC assigned CDI as an urgent threat to the world.[16]

Patients who have failed multiple antibiotic courses for recurrent CDI pose a particular challenge. Fecal microbiota transplantation (FMT) for recurrent CDI has an estimated success rate of 77% to 100% depending on the route of instillation of feces.[89] Therefore, 2018 guidelines recommended FMT for patients with multiple recurrences of CDI who have failed appropriate antibiotic courses.[89]

More recent data, however, demonstrate FMT may not be as effective. A recent meta-analysis of 13 trials showed that studies for recurrent CDI reported higher resolution rates than studies that used FMT for recurrent and refractory CDI. There was a lower efficacy for FMT in patients who have refractory disease compared with standard-of-care antibiotics.[90]

FMT also have safety concerns because the microbiota is associated with a host of unknown chronic medical conditions, including diabetes mellitus, obesity, cancer, and autoimmune disorders. In addition, infectious complications attributed to FMT have included norovirus gastroenteritis, cytomegalovirus, and bacteremias with *Escherichia coli*, *Proteus mirabilis*, *Citrobacter koseri*, and *Enterococcus faecium*.[91] Recently, the Food and Drug Administration issued a warning, after two patients developed extended-spectrum β-lactamase-producing *E coli* infections and one of those patients died.[92] Because of these conflicting reports, more data are needed to determine the overall safety and efficacy of FMT in CDI.

## SUMMARY

HAIs continue to be a major factor in morbidity and mortality, rising health care costs, and the prevalence of drug-resistant organisms. However, most NIs are preventable and effective strategies implemented by the CDC-NHSN in conjunction with health care facilities have demonstrated monitoring and interventions are productive in reducing HAIs to decrease morbidity and mortality, save billions of health care dollars, and reduce the spread of MDROs.

## DISCLOSURE

J-Y. Liu writes infectious disease content for AMPSAA Web site; J.K. Dickter has nothing to report.

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